

Strong latitudinal patterns in the elemental ratios of marine plankton and organic matter

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Supplementary Materials:

Source of Data: All previously published data (upper part of Table S1) and new data (lower part of Table S1) are publically available at several online data repositories or were previously compiled by Sterner et al. (2008; http://www.aslo.org/lo/toc/vol_53/issue_3/1169a1.html). HOT and CMORE-BULA (<http://hahana.soest.hawaii.edu/hot/hot-dogs/>); BATS and Bval37/39 (<http://bats.bios.edu>);
 20 OLIPAC, FLUPAC, DIAPAZON, BIOSOPE, EUMELI, and PROSOPE (<http://www.obs-vlfr.fr/proof/>); NABE, X0804, X0705 and ATP3 (<http://bco-dmo.org>); and Bering Sea (<http://beringsea.eol.ucar.edu/>).

Pigment analysis: We found that the fucoxanthin concentration was significantly positively related to POC, PON, and POP ($p < 0.0001$) as plankton communities with higher diatom abundance had predictably elevated biomass concentration. However, the ratio of the slopes diverged from the Redfield
 25 ratio ($\text{POC}_{\text{slope}}/\text{POP}_{\text{slope}} = 66.4$ and $\text{PON}_{\text{slope}}/\text{POP}_{\text{slope}} = 13.8$). Thus, the C:P and N:P ratios decline in populations with increased fucoxanthin content.

Growth rate analysis: The growth rate hypothesis predicts a strong negative relationship between plankton N:P and C:P ratios and growth rate. As an attempt to test this hypothesis in open ocean communities, we first estimated a “community turnover rate” as a proxy for growth rate. The average
 30 community turnover rate was defined as net primary production divided by the particulate carbon concentration, where both were obtained from our compiled dataset. Sites with low turnover rates generally displayed an overall lower mean value, whereas sites with higher turnover rates had a higher mean ratio and modal distributions (Figure S1C and Table S2). However, turnover rates explained little of the variation in N:P and C:P ratios (Table S4). Changes in temperature have a significant impact on
 35 growth rate and to account for this, most tests of the growth rate hypothesis include a temperature correction¹⁸. However, after correcting for a temperature effect the turnover rate remained a marginal descriptor of N:P and C:P ratios (Table S4). Secondly, we predicted net phytoplankton growth rate using

the global ocean Biogeochemical Element Cycling model³¹(Figure S4). From the model output, we retrieved monthly mean values for net primary production and phytoplankton carbon content for each station and derived the growth rate from the ratio. We also temperature corrected growth rates using a Q_{10} value of 1.5 and 1.88 (e.g., 32-34). This model and other studies showed that phytoplankton growth rate follows a latitudinal gradient with growth rates highest near the equator and declining towards the poles. These patterns were consistent with our observed latitudinal variation in turnover rate and suggested that turnover rate is a reasonable proxy for growth rate. When we compared predicted growth rates with N:P and C:P ratios, we observed a unimodal relationship as expected from the latitudinal gradient of both growth rates and elemental ratios (Figure S4). After a temperature correction, we observed a negative relationship between growth rate and both N:P and C:P ratios (Figure S4). While, the “choice” of temperature correction, Q_{10} of 1.5 vs. 1.88, had a significant impact on the strength of this relationship, at most ~15% of the variation in N:P and C:P ratios was explained. When corrected for differences in temperature, our results provided marginal support for the growth rate hypothesis. Other factors including community composition and nutrient supply rate, were deemed more important in determining the latitudinal patterns observed in the elemental ratios.

Formulation of a two-box model

Here we present the formulation of the 2-box model (Fig. S5A) used to illustrate how variations in the N:P ratio of deep water masses are attenuated in comparison to the the N:P ratio of exported organic matter by the mixing of water masses with different N:P ratios. The model separates the volume of the deep ocean into two boxes of volume V_h and V_l in which the concentrations of nitrate and phosphate are

denoted N_h and P_h for the “high-latitude” box and N_l and P_l for the “low-latitude” box. The exchange of mass between the boxes driven by the large-scale ocean circulation is parameterized using a constant volume flow rate, q .

65 **Phosphate:** With the source terms due to biological export fluxes of phosphorus into the high- and low-latitude boxes denoted by ϕ^h and ϕ^l , the governing equations for the phosphate concentrations are

$$\frac{d}{dt}(P_l) = -\frac{q}{V_l} \cdot (P_l - P_h) + \frac{\phi^l}{V_l} - \gamma P_l, \quad (1a)$$

$$\frac{d}{dt}(P_h) = +\frac{q}{V_h} \cdot (P_l - P_h) + \frac{\phi^h}{V_h} - \gamma P_h, \quad (1b)$$

in which we have added a first order loss term with a rate constant, γ , to balance the source terms and allow the system to reach a steady-state equilibrium. The parameter γ accounts for the transport of
70 nutrient rich water from the deep ocean into the euphotic layer and the removal of inorganic nutrients by phytoplankton.

Nitrate: Denoting the N:P stoichiometric ratios of exported organic matter by $Q_{N:P}^h$ and $Q_{N:P}^l$, the governing equations for nitrate concentrations are

$$\frac{d}{dt}(N_l) = -\frac{q}{V_l} \cdot (N_l - N_h) + \frac{Q_{N:P}^l \phi^l}{V_l} - \gamma N_l, \quad (2a)$$

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$$\frac{d}{dt}(N_h) = +\frac{q}{V_h} \cdot (N_l - N_h) + \frac{Q_{N:P}^h \phi^h}{V_h} - \gamma N_h. \quad (2b)$$

Steady-state solution: The steady-state solution for the concentrations of phosphate and nitrate in the low- and high-latitude boxes are

$$P_l = \frac{1}{\Delta} [q \cdot (\phi^l + \phi^h) + V_h \gamma \phi^l], \quad (3a)$$

$$N_l = \frac{1}{\Delta} [q \cdot (Q_{N:P}^l \phi^l + Q_{N:P}^h \phi^h) + V_h \gamma Q_{N:P}^l \phi^l], \quad (3b)$$

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and

$$P_h = \frac{1}{\Delta} [q \cdot (\phi^l + \phi^h) + V_l \gamma \phi^h], \quad (4a)$$

$$N_h = \frac{1}{\Delta} [q \cdot (Q_{N:P}^l \phi^l + Q_{N:P}^h \phi^h) + V_l \gamma Q_{N:P}^h \phi^h], \quad (4b)$$

in which $\Delta \equiv q\gamma \cdot (V_h + V_l) + \gamma^2(V_h V_l)$.

85 **Seawater N:P ratios:** The N:P ratio for seawater in the low-latitude box is

$$\frac{N_l}{P_l} = \frac{q \cdot (Q_{N:P}^l \phi^l + Q_{N:P}^h \phi^h) + \gamma V_h Q_{N:P}^l \phi^l}{q \cdot (\phi^l + \phi^h) + \gamma V_h \phi^h}, \quad (5)$$

and for the high-latitude box is

$$\frac{N_h}{P_h} = \frac{q \cdot (Q_{N:P}^l \phi^l + Q_{N:P}^h \phi^h) + \gamma V_l Q_{N:P}^h \phi^h}{q \cdot (\phi^l + \phi^h) + \gamma V_l \phi^h}. \quad (6)$$

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Sensitivity: The sensitivity of the solution of the box model can be characterized in terms of three non-dimensional parameters

$$\alpha \equiv \frac{V_l}{V_l + V_h}, \quad \beta = \frac{\phi^l}{\phi^l + \phi^h}, \quad \text{and} \quad \varepsilon \equiv \frac{\gamma V}{q}, \quad (7)$$

95 where $V \equiv V_l + V_h$ denotes the total volume of the deep ocean, and consequently the difference of the N:P of seawater in the two boxes can be expressed as follows

$$\begin{aligned} \frac{N_l}{P_l} - \frac{N_h}{P_h} = & \frac{\beta Q_{N:P}^l + (1 - \beta) Q_{N:P}^h + \beta(1 - \alpha) Q_{N:P}^l \varepsilon}{1 + \beta(1 - \alpha) \varepsilon} \\ & - \frac{\beta Q_{N:P}^l + (1 - \beta) Q_{N:P}^h + \alpha(1 - \beta) Q_{N:P}^h \varepsilon}{1 + \alpha(1 - \beta) \varepsilon}. \end{aligned} \quad (8)$$

The non-dimensional parameter ε can be thought of as the ratio of two timescales: A tracer
100 homogenization timescale controlled by large-scale circulation,

$$\tau = \frac{V}{q} \quad (9)$$

and a bulk residence time for remineralized nutrients in the deep ocean

$$\tau_b = \frac{1}{\gamma}. \quad (10)$$

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A plausible value for the tracer homogenization timescale is $\tau \approx 650$ yrs obtained from a global ocean circulation model³⁶. To determine γ we use the steady-state balance for the phosphate model, which implies

$$\phi = \gamma V \bar{P}, \quad (11)$$

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Where $\bar{P} = (V_l P_l + V_h P_h)/V$ signifies the average phosphate concentration in the model. From the 2009 World Ocean Atlas (32), $V = 1.2 \times 10^{18} \text{ m}^3$ and $\bar{P} = 2.3 \times 10^{-3} \text{ mol/m}^3$ for the ocean volume below 1000 m. Using an estimate of the global net primary production (NPP) of $48.5 \times 10^{15} \text{ gC/yr}$ ³⁷, a globally averaged C:P ratio of 106:1, an average *pe*-ratio of 0.3³⁸, and a power law particulate organic matter flux attenuation profile, ($F(z) = F(z_o)(z/z_o)^{-b}$), with $z_o = 100 \text{ m}$ and $b = 0.9$ ³⁹, we estimate the export flux at 1000 m depth from

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$$\phi \approx \frac{pe \cdot NPP}{12 \cdot 106} (0.1)^{0.9} = 1.4 \times 10^{12} \text{ mol P/m}^3, \quad (12)$$

so that $\gamma \approx 2000$ yrs, and

$$\varepsilon = 0.3. \quad (13)$$

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We can simplify Eq. (8) if $\varepsilon \ll 1$ and derive the following expression for the N:P ratio of seawater in the two respective boxes,

$$\frac{N_l}{P_l} - \frac{N_h}{P_h} = (\beta - \beta^2)(Q_{N:P}^l - Q_{N:P}^h)\varepsilon + \mathcal{O}(\varepsilon^2), \quad (14)$$

125 provided $\varepsilon \ll 1$. Thus a rough rule of thumb is that the difference in the N:P ratio between the boxes is reduced by a factor of $\varepsilon/4$ compared to the difference in the N:P ratio of the biological export flux.

Figure S5b shows the difference in the seawater N:P ratio between the low- and high-latitude boxes plotted as a function of the difference in the N:P ratio of exported organic matter for the reference
130 parameter values, $\alpha = 0.5$, $\beta = 0.5$, and $\varepsilon = 0.325$. The small value of the slope, ~ 0.08 is in good agreement with the rule of thumb value of 0.25ε appropriate for $\beta = 0.5$.

Figure S6 shows contour plots of the difference in the N:P ratio of seawater between the low- and high-latitude boxes as a function of the difference in the N:P ratio of exported organic matter from the low-
135 and high-latitude boxes and as a function of the non-dimensional parameters α , β , and ε . In accord with the asymptotic formula in Eq. (14), the difference of the N:P ratio of seawater in the two boxes shows very little dependence on α (c.f. Fig S6a). Fig. S6b shows that the largest differences in the N:P ratio of seawater occur for the case of $\beta = 0.5$. As β increases towards unity, a progressively larger proportion of the exported organic matter is exported with the same N:P ratio which decreases the ability of plankton
140 N:P variations to be imprinted in the deep water masses. Only in the limit of $\varepsilon \rightarrow 1$ when the bulk residence time of exported nutrients becomes comparable to the timescale with which the circulation homogenizes tracer gradients do we find any significant variations in the N:P of deep water masses (c.f. Fig S6c). However, even for $\varepsilon = 1$, the seawater N:P ratio differences between the low- and high-latitude boxes are attenuated by a factor of ~ 5 compared to the difference in the N:P ratio of exported organic
145 matter.

Figures and Captions:

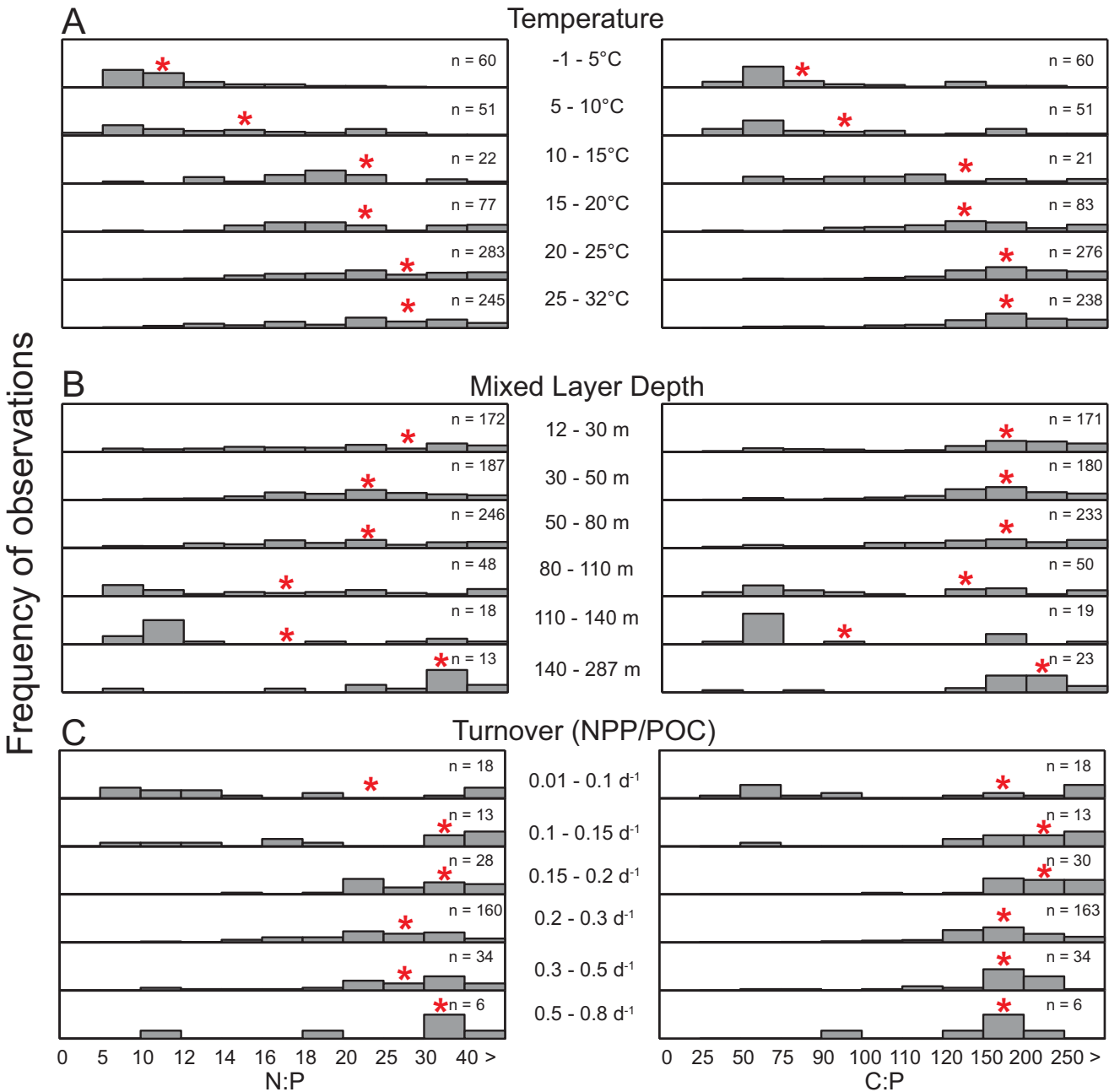


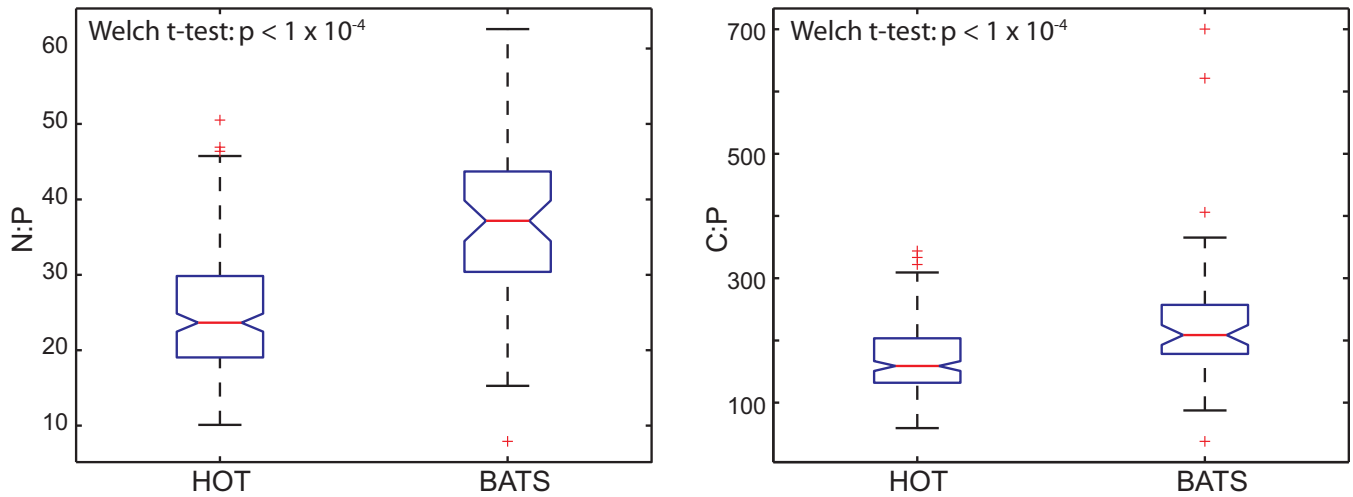
Figure S1: **Elemental stoichiometry of marine environments.** Each histogram displays the frequency

of stations (0 – 100%) with a N:P or C:P ratio within the specified range. The red * specifies bin

containing the lognormal mean of each ratio and n specifies the number of stations in each histogram. **A.**

The N:P and C:P ratio binned by temperatures. **B.** The N:P and C:P ratio binned by turnover rates. The

turnover rate is defined as the net primary productivity divided by the particulate organic carbon concentration. C. The N:P and C:P ratio binned by mixed layer depth. The temperature and turnover rates are the average values in the top 50 m at a specific station. Please note the unequal bin size in the histogram.



160 Figure S2: **Plankton elemental stoichiometry at the Hawaii Ocean Time-series (HOT, n = 199) and**
Bermuda Atlantic Ocean Time-series Study (BATS, n = 60). **A.** The N:P ratio and **B.** the C:P ratio.
The top and bottom of each box plot represents the 25th and 75th percentile and the line in the middle
represents the median. Notches display the variability of the median between samples. We used the
Welch t-test to account for uneven variance at the two sites.

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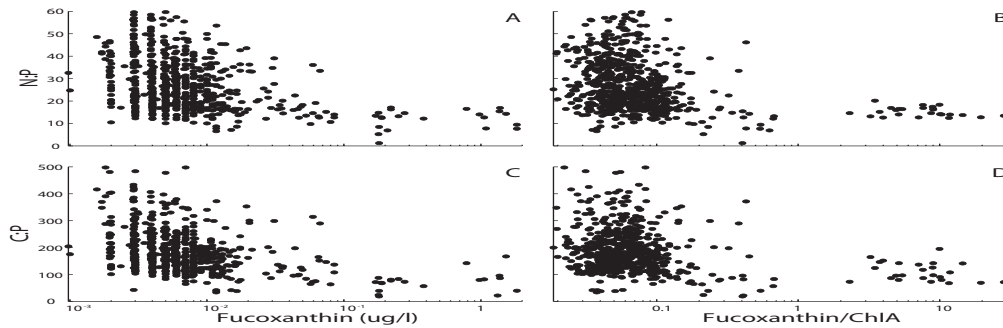


Figure S3: **Relationship between fucoxanthin concentration and elemental ratios.** **A.** The

170 relationship between fucoxanthin and the N:P ratio ($n=746$, $p_{\text{spearman}} = 1 \times 10^{-22}$). **B.** The relationship
between fucoxanthin normalized to chlorophyll A concentration and the N:P ratio ($n=746$, $p_{\text{spearman}} = 4 \times$
10⁻¹⁸). **C.** The relationship between fucoxanthin and the C:P ratio ($n=695$, $p_{\text{spearman}} = 2 \times 10^{-25}$). **D.** The
relationship between fucoxanthin normalized to chlorophyll A concentration and the N:P ratio ($n=695$,
 $p_{\text{spearman}} = 1 \times 10^{-7}$). The pigment fucoxanthin is commonly found in diatoms and can be considered a
175 broad proxy for diatom abundance²⁴. Note that the datapoints represent specific observations and not
station averages.

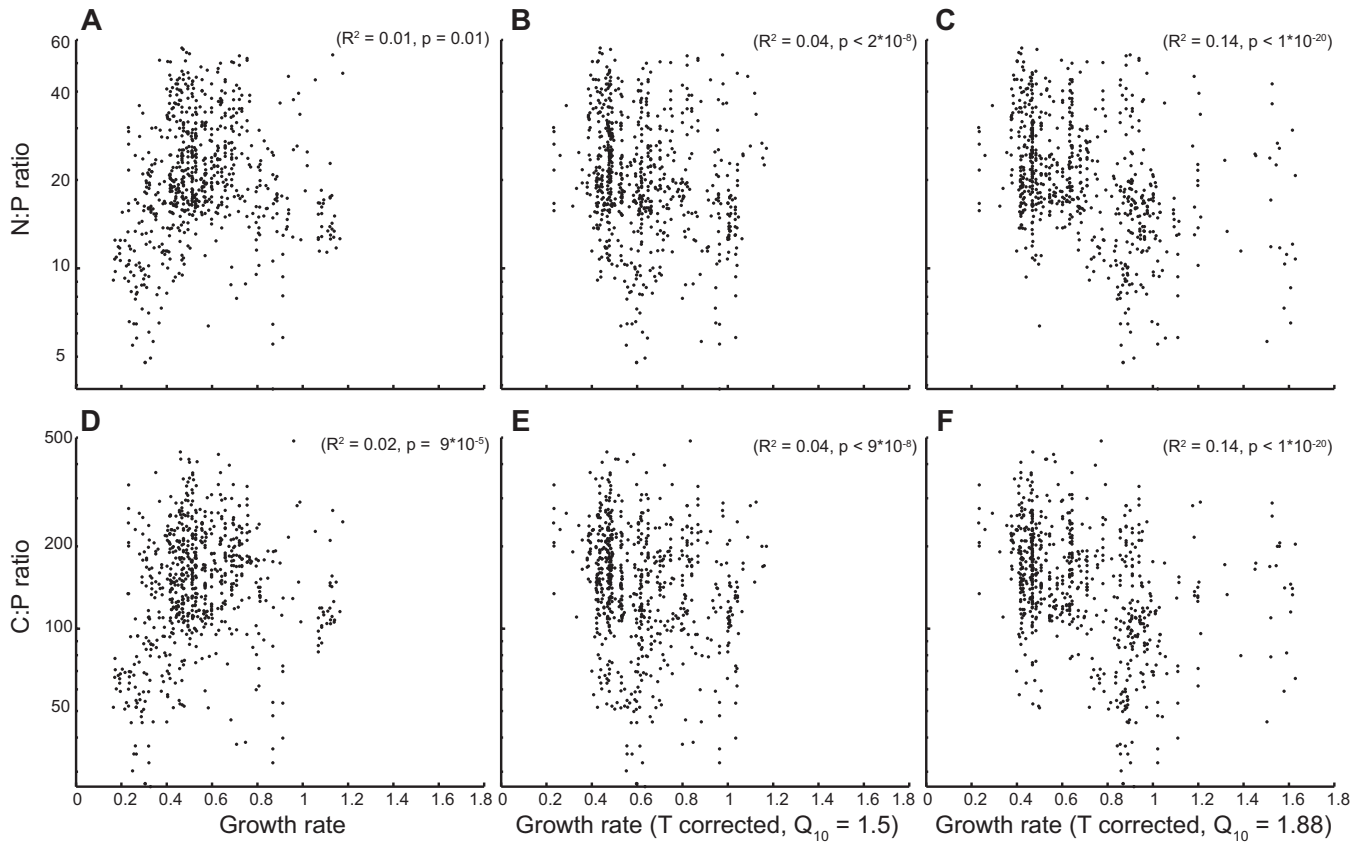


Figure S4. **Relationship between predicted phytoplankton growth rates, temperature, and**

180 **community elemental stoichiometry.** Phytoplankton growth rates were predicted in a global ocean biogeochemical model³¹. **A and D.** Relationship between N:P and C:P ratios and phytoplankton growth rates. **B and E.** Relationship between N:P and C:P ratios and phytoplankton growth rates corrected for temperature using $Q_{10} = 1.5$. **C and F.** Relationship between N:P and C:P ratios and phytoplankton growth rates corrected for temperature using $Q_{10} = 1.88$ (e.g.,^{32,33,35}). Significance values were based on

185 linear regression of log transformed elemental ratios.

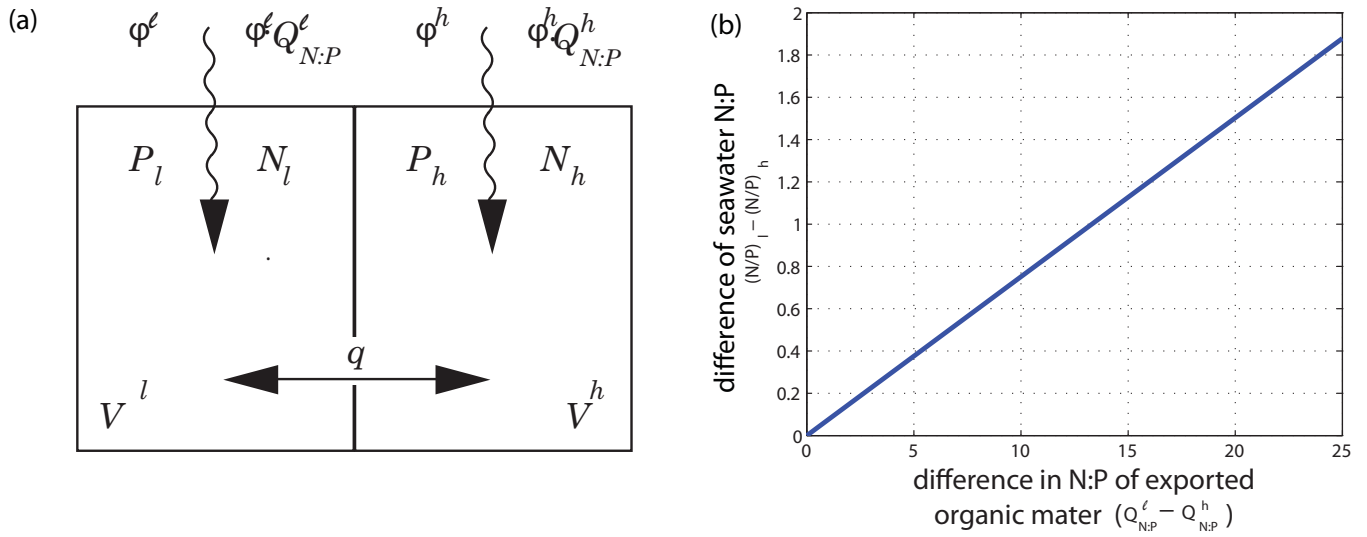
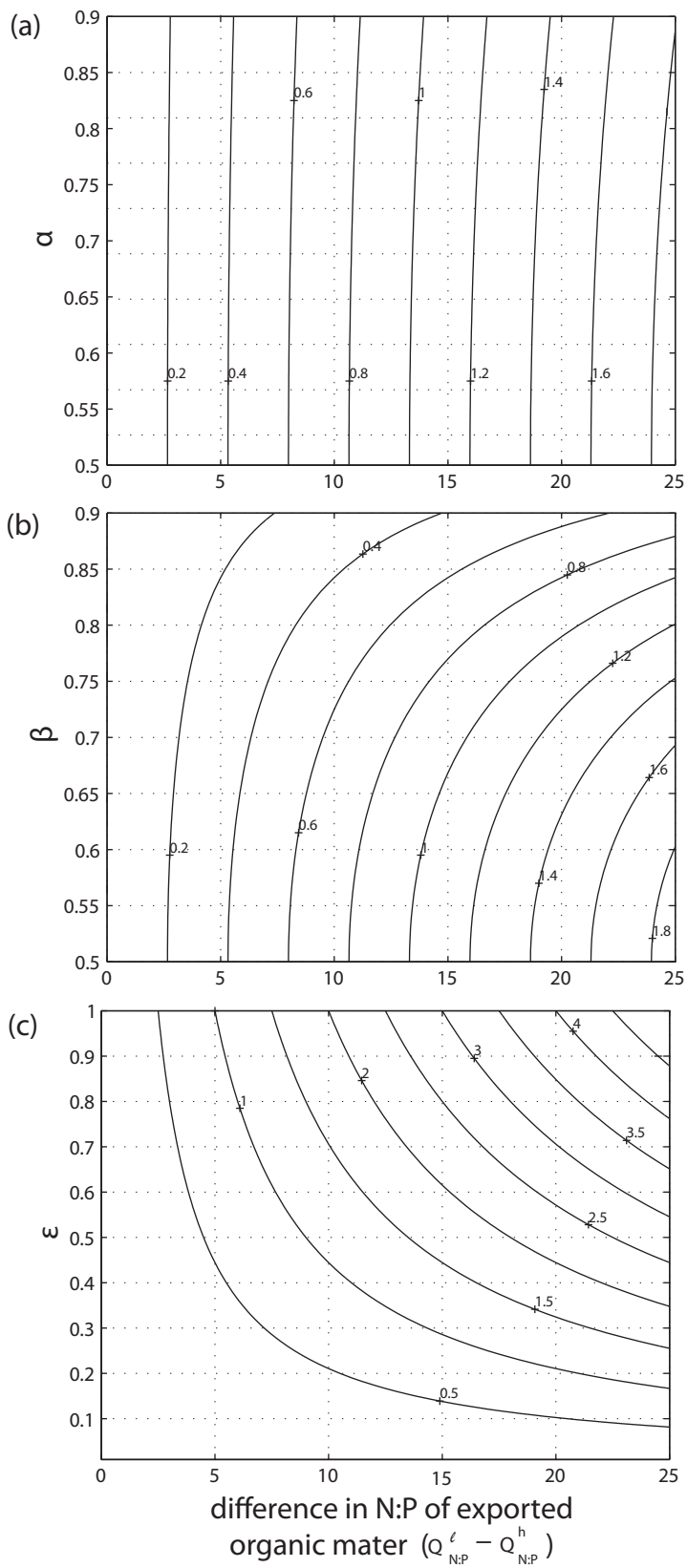


Figure S5: (a) **Schematic diagram illustrating the formulation of the two-box model.** (b) Difference in the N:P ratio of seawater as a function of the difference in the N:P ratio of the exported organic matter with fixed $\alpha = 0.5$, $\beta = 0.5$, and $\varepsilon = 0.325$.



195 **Figure S6: Contour plots of the difference in the N:P ratio of seawater as a function of $Q_{\text{N:P}}^{\ell} - Q_{\text{N:P}}^{\text{h}}$**
and the difference in the N:P exported organic matter. (a) α , the ratio of the volume of the low-
latitude box to the total volume, with fixed $\varepsilon = 0.325$ and $\beta = 0.5$. (b) β , the ratio of the export flux of
phosphorus to the total export flux, with fixed $\varepsilon = 0.325$ and $\alpha = 0.5$, and (c) ε , the ratio of the tracer
homogenization timescale by the large-scale circulation to the bulk residence time of remineralized
200 nutrients in the deep ocean, with fixed $\alpha = 0.5$ and $\beta = 0.5$.

Table S1: Summary of datasets including cruises and time-series used herein. The bottom part of the table lists samples collected as part of this study.

Dataset	Year	Longitude	Latitude	Stations		Samples			References
				N:P	C:P	P	N	C	
HOT	1989 - 2009	158W	22N	199	200	1575	1630	1630	¹⁰
BATS	1995 - 2010	64W	31N	60	60	748	708	802	¹²
OLIPAC	1994	150W	16S - 1N	13	0	120	152	0	⁴⁰
FLUPAC	1994	149W - 179W	14S - 6N	34	34	400	375	374	⁴¹
SEEDS	2001	164E - 166E	49N	6	6	57	129	129	⁴²
DIAPAZON	2001 - 03	166E - 167E	24S - 20S	28	28	303	203	202	⁴³
CMORE-BULA	2007	159W - 170W	16S - 17N	7	7	48	45	45	⁴⁴
BIOSCOPE	2004	72W - 141W	35S - 8S	21	22	164	156	162	⁴⁵
EUMELI	1991 - 92	21W - 31W	18N - 21N	28	0	323	318	0	⁴⁶
PROSCOPE	1999	10W - 22E	30N - 43N	23	23	223	220	217	⁴⁷
NABE	1989	21W - 30W	18N - 34N	20	20	239	236	237	⁴⁸
MOOGLI	1997 - 1999	5W - 5W	43N - 43N	11	10	123	121	114	⁴⁹
MD03/ICHTYO	1974	26E - 78E	56S - 24S	168	169	169	169	168	⁵⁰
Pulse-26	1995 - 96	121W - 176W	54S - 36N	5	5	31	31	31	⁵¹
Tuamotu	1985 - 96	141W - 148W	15S - 18S	14	14	14	16	16	⁵²
Panama Basin	1976	86E	1N	0	1	3	0	3	⁵³
Baltic Sea	1999	16E - 22E	56N - 60N	7	7	7	7	7	⁵⁴
East Atlantic	1973	10E	31N	4	4	19	19	19	⁵⁵
Bering Sea	2008 - 2010	161W - 179W	54N - 63N	28	28	296	272	291	This study
Bval37 and 39	2006 - 07	64W - 66W	20N - 34N	1	1	181	50	50	This study
X0804	2008	45W - 64W	20N - 32N	20	20	401	222	227	This study
X0705	2007	56W - 65W	27N - 38N	24	24	238	215	215	This study
ATP3	2006	64W - 66W	21N - 32N	13	13	130	129	129	This study
Sum				742	733	5631	5318	4965	

Table S2: Mean N:P and C:P ratios in different sample sets. The 2.5% and 97.5% confidence intervals were estimated by assuming a lognormal distribution of each sample bin. The number of predicted Gaussian distributions test to determine if a given sample set can best be represented by a single mean value.

Bin	mean N:P	Ci _{2.5%}	Ci _{97.5%}	Predicted Gaussian distributions	mean C:P	Ci _{2.5%}	Ci _{97.5%}	Predicted Gaussian distributions
All	23.4	7.5	56.0	5	166.6	51.0	409.9	1
Latitude								
60N	14.5	5.2	32.4	5	119.6	42.6	269.2	1
50N	17.2	9.4	29.1	N/A ^a	125.1	62.3	225.5	N/A ^a
40N	25.4	12.6	45.9	5	192.1	85.4	374.7	1
30N	36.6	9.4	99.1	1	225.7	71.1	546.7	1
20N	24.9	9.6	53.2	1	171.0	67.5	360.7	1
10N	33.5	5.4	113.8	N/A ^a	188.0	70.9	408.5	N/A ^a
0	18.0	7.4	37.0	1	136.9	62.5	262.2	1
10S	20.6	10.2	37.4	1	216.0	82.2	466.5	4
20S	26.1	11.4	51.5	1	226.1	85.0	492.2	5
30S	17.7	13.2	23.2	5	134.3	82.5	206.9	1
40S	17.5	10.4	27.6	1	107.0	58.9	179.2	5
50S	12.5	4.8	26.7	1	74.0	26.6	165.5	1
60S	10.6	8.3	13.3	N/A ^a	65.5	47.5	88.1	N/A ^a
Nitrate								
<0.003	24.7	13.3	42.1	5	170.4	95.7	281.4	1
<0.03	28.0	10.3	61.7	1	195.0	78.1	407.5	1
<0.3	26.2	8.0	64.7	1	186.3	62.7	434.1	1
<3	21.0	8.1	44.8	1	156.6	53.1	363.0	1
<15	16.2	6.8	32.8	1	107.6	54.9	190.9	1
<27	12.8	5.2	26.5	5	78.4	27.7	177.3	1

Temperature								
<5	11.6	5.5	21.6	1	84.7	33.1	179.6	1
<10	15.5	5.1	36.5	1	100.0	26.5	266.7	1
<15	20.5	9.6	38.6	1	128.5	49.8	274.0	1
<20	20.7	10.1	37.7	5	139.4	63.9	266.5	1
<25	26.1	8.6	61.6	1	182.8	64.3	414.8	1
<32	25.6	9.5	56.0	1	190.1	72.6	409.9	1
Mixed Layer depth								
30	27.8	6.4	79.4	1	198.8	54.1	522.8	1
50	23.7	10.7	45.8	1	173.6	70.8	358.5	1
80	21.7	7.5	49.5	1	152.4	48.9	365.2	1
110	17.0	5.7	39.6	1	111.3	34.2	273.5	1
140	16.5	5.0	40.9	1	97.5	28.6	245.9	1
287	31.4	11.1	71.0	1	202.7	72.7	454.1	1
Turnover rate								
0.1	21.5	4.8	62.7	1	198.6	28.5	706.8	1
0.15	30.9	7.4	87.0	1	211.7	75.9	474.3	1
0.2	30.7	16.0	53.5	1	219.6	133.8	340.6	1
0.3	25.9	12.3	48.4	1	174.7	85.4	319.1	1
0.5	29.9	12.2	61.7	1	172.4	86.0	310.5	1
0.8	30.8	9.0	77.8	N/A ^a	158.1	93.4	251.1	N/A ^a

Table S3: The lognormal mean N:P and C:P ratios for all samples and in latitudinal bins as estimated after corrected for spatial autocorrelation. To correct for spatial autocorrelation, we first estimated the mean N:P and C:P values for each 1° grid cell before we calculated the mean of a specific bin.

Bin	mean N:P	mean C:P
Obs	303	291
All	19.0	131.7
Weighted mean	20.6	146.9
Latitude		
60	13.9	116.2
50	16.2	114.0
40	27.6	197.9
30	42.0	236.9
20	30.2	173.1
10	42.0	212.8
0	19.3	145.4
-10	21.3	223.0
-20	25.7	214.0
-30	17.5	136.8
-40	17.2	104.3
-50	11.9	71.1
-60	10.4	63.4

Table S4: Pearson and Spearman correlation analysis between environmental variables and the N:P and C:P ratios. The N:P and C:P ratios were log transformed before the analysis. The turnover rate was also temperature corrected using $Q_{10} = 1.5$.

Variables	Pearson				Spearman			
	N:P		C:P		N:P		C:P	
	R ²	p	R ²	p	R ²	p	R ²	p
Nitrate (log)	-0.33	< 1x10 ⁻⁴	-0.38	< 1x10 ⁻⁴	-0.42	< 1x10 ⁻⁴	-0.41	< 1x10 ⁻⁴
Temperature	0.36	< 1x10 ⁻⁴	0.42	< 1x10 ⁻⁴	0.27	< 1x10 ⁻⁴	0.35	< 1x10 ⁻⁴
Turnover rate	0.14	0.025	-0.030	0.62	0.033	0.60	-0.19	0.0018
Turnover (T corr.)	0.002	0.976	-0.052	0.40	0.01	0.870	-0.191	0.002
Mixed layer depth	-0.067	0.078	-0.13	0.0004	-0.10	0.008	-0.18	< 1x10 ⁻⁴

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Table S5: Summary of *Prochlorococcus*, *Synechococcus*, and eukaryotic pico- and nano-phytoplankton specific carbon, nitrogen, and phosphorus quotas (fg cell⁻¹), population-specific carbon, nitrogen, and phosphorus content (μmol l⁻¹, except phosphorus, nmol l⁻¹) and percent contributions to whole community particulate carbon, nitrogen, and phosphorus. We also counted heterotropic bacteria but used a previously published estimate for cell quota to facilitate a mass balance.

Population	Station	Depth	Cells	C quota	N quota	P quota	C content		N content		P content	
		(m)	(cells ml ⁻¹)	(fg cell ⁻¹)	(fg cell ⁻¹)	(fg cell ⁻¹)	(μmol l ⁻¹)	(% whole)	(μmol l ⁻¹)	(% whole)	(nmol l ⁻¹) ^a	(% whole)
<i>Prochlorococcus</i>	31°40'N	1	54,239	49	10.2	0.90	0.22	13.1	0.040	12.1	1.6	16.5
	(BATS)	30	119,574	70	11.5	1.25	0.70	33.8	0.098	26.7	4.8	40.6
		100	47,848	90	14.5	1.85	0.36	22.7	0.050	16.2	2.9	28.6
	23°40'N	1	51,653	30	6.0	0.25	0.13	6.2	0.022	7.0	0.4	5.2
		30	51,975	50	7.9	0.38	0.22	11.1	0.029	7.1	0.6	8.0
		100	96,247	60	9.9	0.45	0.48	25.7	0.068	19.1	1.4	15.5
				58±20	10.0±2.9	0.85±0.62	0.35±0.21	18.8±11.2	0.051±0.028	15.0±8.3	2.0±2.0	20.7±17.5
<i>Synechococcus</i>	31°40'N	1	10,965	313	61.5	6.8	0.29	16.9	0.048	14.7	2.4	25.2
	(BATS)	30	9,626	350	65.5	8.00	0.28	13.6	0.045	12.2	2.5	20.9
		100	271	390	67.5	9.00	0.01	0.6	0.001	0.4	0.1	0.8
	23°40'N	1	10,828	263	45.0	3.45	0.24	11.2	0.035	11.0	1.2	15.1
		30	2,410	300	56.0	4.05	0.06	3.1	0.010	2.3	0.3	3.9
		100	4,891	310	53.0	3.40	0.13	6.8	0.019	5.2	0.5	6.0
				321±44	58.1±8.4	5.78±2.47	0.17±0.12	8.9±6.3	0.026±0.019	14.4±5.3	2.0±1.0	12.5±11.2
Eukaryotes	31°40'N	1	631	8,418	1,500	200	0.44	26.2	0.068	20.6	4.1	42.7
	(BATS)	30	775	5,500	901	145	0.36	17.2	0.050	13.6	3.6	30.6
		100	1,059	5,000	775	125	0.44	27.9	0.059	19.2	4.3	42.8
	23°40'N	1	495	8,317	1,600	175	0.34	16.2	0.057	17.9	2.8	35.0
		30	222	5,450	1,005	135	0.10	5.2	0.016	3.9	1.0	12.1
		100	745	5,300	850	130	0.33	17.6	0.045	12.7	3.1	34.7
				6,330±	1,105±							
				1,586	354							
						152±29.7	0.34±0.12	17.9±6.7	0.048±0.019	14.4±5.3	3.0±1.0	33.5±12.8

H. Bacteria	31°40'N	1	499,175	7.1 ^b	1.4 ^b	0.22 ^b	0.30	17.5	0.048	14.8	3.5	37.1
	(BATS)	30	344,400	7.1	1.4	0.22	0.20	9.9	0.033	9.1	2.4	20.6
		100	461,250	7.1	1.4	0.22	0.27	17.3	0.045	14.7	3.3	32.7
	23°40'N	1	387,450	7.1	1.4	0.22	0.23	10.8	0.038	11.9	2.7	34.4
		30	430,500	7.1	1.4	0.22	0.26	13.1	0.042	10.1	3.1	38.2
		100	506,350	7.1	1.4	0.22	0.30	16.1	0.049	13.8	3.6	39.9
				-----	-----	-----	0.26±0.04	13.9±2.0	0.043±0.001	12.5±1.8	3.0±1.0	33.1±4.8
	Sum of populations	31°40'N	1	-----	-----	-----	1.25	73.8	0.204	62.1	11.6	121.5
		(BATS)	30	-----	-----	-----	1.54	74.6	0.227	61.6	13.4	112.7
Sum of populations		100		-----	-----	-----	1.08	68.5	0.154	50.6	10.5	104.9
	23°40'N	1		-----	-----	-----	0.94	44.4	0.151	47.9	7.2	89.6
		30		-----	-----	-----	0.63	32.6	0.097	23.5	5.0	62.2
		100		-----	-----	-----	1.24	66.2	0.181	50.8	8.7	96.1
				-----	-----	-----	1.11±0.3	60±17	0.17±0.05	49±14	9.0±3.0	98±21
Whole community	31°40'N	1		-----	-----	-----	1.69	-----	0.328	-----	9.5	-----
	(BATS)	30		-----	-----	-----	2.06	-----	0.368	-----	11.9	-----
		100		-----	-----	-----	1.58	-----	0.305	-----	10.0	-----
	23°40'N	1		-----	-----	-----	2.12	-----	0.316	-----	8.0	-----
		30		-----	-----	-----	1.94	-----	0.412	-----	8.0	-----
		100		-----	-----	-----	1.87	-----	0.356	-----	9.0	-----
				-----	-----	-----						
				-----	-----	-----						
				-----	-----	-----						

^a Note change in units to nmol l⁻¹.

^b Carbon, nitrogen and phosphorus content calculated from average biovolume and element/biovolume

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